





Management and Curation of EarthPortal Resources

How the Java code implements the O'FAIRe questions

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1 Introduction

In this short report, our goal is to look at how the Java code of O'FAIRe implements the 61 questions presented in Report 2. The JSON file says what is asked and how many points an answer is worth, but it is the code that decides which answer an artefact receives. Since the scores shown in EarthPortal come from this code, I wanted to read it once, question by question.

2 Development

2.1 Where the code is, and what I read

The source code of O'FAIRe is public[1]. I read it as it was on 4 June 2026 (branch `master`, commit `24def6b`).

Each sub-principle has one Java class (`F1.java`, `F2.java`, and so on until `R13.java`), in the folder `principles/criterion/impl`. For one artefact, the code asks the portal API for the latest submission with all its fields, and then goes through the questions one by one. For most questions, the code only checks that the field has a value; I then say that the field is *filled*. Whether the value is correct is not checked, except where Table 2 says so.

The points of the questions are added for each sub-principle, then for each principle, then for the artefact. The score out of 100 is the total divided by 478, and the decimals are dropped (for example, 77.9 is shown as 77).

2.2 Four kinds of questions

The code does not treat the 61 questions in the same way. Table 1 gives the four cases.

Kind	What the code does	Questions	Points
Tested	It reads one or several fields of the artefact and chooses the answer.	38	300
Given	It gives the best answer to every artefact, because the answer depends on the portal and not on the artefact.	12	106
Not implemented	It gives 0 to every artefact, with the message *This question cannot be evaluated in the repository at this time.*	10	70
Always 0	F3Q1 only: it gives 0 to every artefact, with an unfinished message.	1	21
Total		**61**	**497**

Table 1: The four kinds of questions in the code. *Points* is the sum of the best answers.

2.3 Question by question

Table 2 gives the rule of each question in one or two lines. The fields are written with the names the portal uses in its API; Report 3 gives the MOD name of each of them. A *valid URL* is a text that has the form of an `http` or `https` address. A URL *resolves* when the connection is made within one second and the server answers with the code 200, or with the code 302 and a content type; redirections are not followed.

[1] `https://github.com/agroportal/fairness`, folder `src/main/java/fr/lirmm/fairness/assessment`.

Table 2: What the code does for each question. *Max* is the best answer of the question.

Question	Max	Kind	What the code does
Findable			
F1Q1	9	Tested	`URI`: empty 0; filled but not a valid URL 3; a valid URL 9.
F1Q2	11	Tested	`identifier`: empty 0; not a valid URL 3; a valid URL 6; a `doi.org` address that the DOI service does not know 9; a DOI that the DOI service knows 11. With several identifiers, the last one decides.
F1Q3	12	Given	
F1Q4	9	Tested	`versionIRI`: empty 0; not a valid URL 2; a valid URL that does not resolve 4; a URL that resolves 9.
F2Q1	16	Tested	Counts the filled fields among 25: the 24 of the list in Report 3, and `translator`. 2 points for each, up to 8 fields.
F2Q2	4	Tested	Counts the filled values among six: the link to the metrics, the numbers of classes, properties, individuals and axioms, and `preferredNamespacePrefix`. The points are 0, 1, 2, 3, 3, 4, 4.
F2Q3	7	Tested	Counts the filled fields among the 34 behind the list in Report 3. 1 point for each, up to 7 fields.
F3Q1	21	Always 0	The message is *This question*, and nothing more.
F3Q2	11	Given	
F3Q3	10	Given	
F4Q1	6	Tested	Starts at 1 if the portal itself is in the list of repositories of O'FAIRe. Then, for each value of `includedInDataCatalog`, adds 1 if it is in the list of repositories and 1 if it is in the list of libraries. 2 points for each unit, up to 3.
F4Q2	10	Tested	The same count, with the list of repositories only. 2 points for each unit, up to 5.
F4Q3	8	Not implemented	
Accessible			
A1Q1	6	Tested	Tests whether `URI` and each `identifier` is a valid URL that resolves. One of them is enough for the 6 points.
A1Q2	7	Given	
A1Q3	24	Tested	Asks the `URI`, and then the address of the metadata in the portal API, for five formats (JSON, XML, HTML, plain text, RDF/XML), and counts the formats accepted. 3 points for each, up to 8.
A1Q4	6	Tested	`endpoint` filled: 6. The endpoint itself is not tested.
A1.1Q1	20	Given	
A1.1Q2	4	Given	
A1.1Q3	4	Given	
A1.2Q1	11	Given	
A1.2Q2	11	Given	
A2Q1	7	Given	
A2Q2	5	Given	
A2Q3	4	Given	
A2Q4	4	Tested	Counts `status` and `deprecated`. 2 points for each.
Interoperable			
I1Q1	18	Tested	`hasOntologyLanguage` must be exactly PDF, TXT, CSV, XML, OBO, RDFS, SKOS or OWL, for 5, 7, 9, 10, 11, 12, 15 or 18 points. Any other value: 0.
I1Q2	10	Tested	`hasOntologyLanguage` is exactly XML, RDFS, SKOS or OWL: 10.
I1Q3	5	Tested	`hasOntologySyntax` filled: 5.
I1Q4	5	Tested	`hasFormalityLevel` filled: 5.
I1Q5	4	Tested	`hasFormat` and `isFormatOf` both filled: 4. One of them alone: 0.

Question	Max	Kind	What the code does
I2Q1	5	Tested	`useImports` filled: 5.
I2Q2	5	Tested	`ontologyRelatedTo` filled: 5.
I2Q3	3	Not implemented	
I2Q4	5	Tested	`isAlignedTo` filled: 5.
I2Q5	7	Not implemented	
I2Q6	2	Tested	At least one of `similarTo`, `translationOfWork`, `explanationEvolution`, `generalizes` and `viewOf` filled: 2.
I2Q7	5	Tested	O'FAIRe has a list of 34 metadata vocabularies, each with a weight of 0.25, 0.5 or 0.75. The result is 5 times the average weight of the vocabularies of `metadataVoc` that are in this list.
I3Q1	20	Not implemented	
I3Q2	6	Not implemented	
I3Q3	7	Tested	Looks at the fields of the list in Report 3 and at `naturalLanguage`, and counts those whose values are all valid URLs. 1 point for each, up to 7.
Reusable			
R1Q1	5	Tested	Counts `prefLabelProperty`, `synonymProperty`, `definitionProperty`, `authorProperty` and `obsoleteProperty`. 1 point for each.
R1Q2	3	Tested	Counts `hierarchyProperty`, `obsoleteParent` and the metric `maxDepth`. 1 point for each.
R1Q3	7	Not implemented	
R1Q4	6	Not implemented	
R1Q5	6	Not implemented	
R1Q6	5	Not implemented	
R1.1Q1	15	Tested	`hasLicense`: empty 0; not a valid URL 4; a valid URL that does not resolve 8; a URL that resolves but accepts none of the five formats of A1Q3 12; a URL that accepts at least one of them 15.
R1.1Q2	7	Tested	`viewingRestriction` filled: 4; `morePermissions` filled as well: 7.
R1.1Q3	15	Tested	Counts `useGuidelines` and `contact`: 0, 7 or 15. The copyright holder of the question is read from the contacts.
R1.2Q1	8	Tested	Counts `hasCreator`, `curatedBy`, `hasContributor` and `translator`: 0, 3, 5, 7 or 8.
R1.2Q2	6	Tested	Counts `source`, `wasGeneratedBy` and `wasInvalidatedBy`. 2 points for each.
R1.2Q3	6	Tested	Counts `accrualMethod`, `accrualPeriodicity` and `accrualPolicy`. 2 points for each.
R1.2Q4	4	Tested	Counts `version`, `hasPriorVersion` and the link to the list of submissions. 2 points for each, up to 2.
R1.2Q5	2	Not implemented	
R1.2Q6	6	Tested	Counts `usedOntologyEngineeringTool`, `usedOntologyEngineeringMethodology` and `conformsToKnowledgeRepresentationParadigm`. 2 points for each.
R1.2Q7	4	Tested	Counts `designedForOntologyTask` and `competencyQuestion`. 2 points for each.
R1.2Q8	2	Tested	`fundedBy` filled: 2.
R1.3Q1	10	Tested	Counts `endorsedBy` and the projects that use the artefact in the portal. 5 points for each.
R1.3Q2	20	Tested	Looks at the groups of the artefact: 10 if a group name contains WHEAT, CROP or INRAE; 15 if it contains OBO-FOUNDRY; 20 if, in addition, the acronym is BFO, CHEBI, GO, PATO, PO or PR.
R1.3Q3	6	Tested	`viewingRestriction` contains the word *public*: 6.

3 Conclusion

Throught the reading of the code, I noticed a few points that are not obvious from the JSON file.

- **The lists of properties** in the JSON file does not decide the score, it only decides what is displayed next to the score.
  Actually, in `questions.config.json` (Report 2), most questions carry a properties list. For example, I1Q5 lists `dct:hasFormat` and `dct:isFormatOf`. Reading this, We would think that the program takes this list and checks those properties to compute the score. However, the code works differently,in two separate steps:
  - **The test**. Each question has its own small piece of Java, and the fields to check are written by hand inside it. For I1Q5, the code says, in substance, "if `hasFormat` is filled and `isFormatOf` is filled, give 4 points, otherwise 0". This step never looks at the JSON list.
  - **The display**. Once the score is chosen, another piece of code (`Tester.java`) takes the JSON list and copies the values of those properties into the result. This is what we see in the O'FAIRe panel, next to the explanation. For **OSO**, it shows the six formats under `dct:hasFormat` and *null* under `dct:isFormatOf`.
    So there are two lists for each question: one in the Java code, which gives the points, and one in the JSON, which only shows values to the reader. Most of the time the two lists are the same, but they are maintained separately, and in three questions they have drifted apart:
    - F2Q1: the code counts the `translator`, but the JSON list does not show it.
    - I3Q3: the code also checks the `naturalLanguage`, which the JSON list does not show. The JSON list shows the subjects, but the code can never count them.
    - R1.3Q2: the code also looks at the `acronym`, which is not in the JSON list.
- **F3.** F3Q1 always gives 0, F3Q2 and F3Q3 are always given, and the weight of F3 (21) counts only these two. Every artefact therefore has 21 out of 21 on F3.
- **I1.** The best answers of I1 add up to 42, while its weight is 44. No artefact can have more than 95 % on I1.
- **I2Q7.** The highest weight in the list of vocabularies is 0.75, so the result cannot be more than 3.75 out of 5.
- **A1Q1.** The answer worth 3 points is never used: one identifier that resolves already gives 6.
- **R1.3Q2.** The names of the groups are written in the code. None of the groups used in EarthPortal has such a name, so its artefacts have 0 out of 20.
- **F4Q1 and F4Q2.** EarthPortal is not in the list of repositories of O'FAIRe. An artefact that declares only EarthPortal in `includedInDataCatalog` has 0 on both questions.

For an artefact of EarthPortal, this leaves 386 of the 478 points within reach: 478, minus the 70 points of the ten questions that are not implemented, minus the 20 points of R1.3Q2, minus the 2 points that no question of I1 gives.